

YU FENG

Ph.D. Candidate | Imaging Systems

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PROFESSIONAL SUMMARY

Trilingual researcher (English, Japanese, Chinese) with 5+ years of experience in LSI design, hardware/software implementation, and performance characterization. Specializes in multi-tap CMOS Image Sensor (CIS) systems development, achieving programmable 56-126 dB single-exposure HDR with motion/flicker mitigation. Additionally developed robust biomedical imaging systems resilient to ambient light and motion artifacts. Proven track record of full-cycle tape-out process (180 nm), FPGA prototyping, and first-author peer-reviewed publications.

TECHNICAL SKILLS

IC Design & Verification: Cadence Virtuoso (Schematic/Layout), Spectre (Simulation), Calibre (DRC/LVS), PCB design, full-cycle tape-out (180 nm CIS process).

Characterization: Dynamic Range (DR)/SNR analysis, Linearity tuning, Full Well Capacity (FWC) optimization, Crosstalk suppression.

Hardware & Architecture: Multi-tap CIS architecture, FPGA (Verilog, Quartus, ModelSim).

Software: MATLAB (Image processing, simulation), Python, C/C++.

Languages: English (TOEIC 990/990), Japanese (JLPT N1), Chinese (Native).

EDUCATION

Ph.D. Shizuoka University Hamamatsu, Japan Nanovision Technology Advisor: Prof. Keiichiro Kagawa	2023.10 – 2026.09 (Expected)
M.Eng. Shizuoka University Hamamatsu, Japan Electronics Engineering	2021.10 – 2023.09
B.Eng. Shizuoka University Hamamatsu, Japan Electrical and Electronic Engineering	2016.04 – 2020.03

CORE RESEARCH EXPERIENCE

High Dynamic Range (HDR) Imaging System Development	2023 – Present
- Systematically optimized reset and transfer gate voltages to suppress residual charge crosstalk, achieving a programmable 56–126 dB Single-Exposure HDR. - Developed a programmable charge-splitting method that mitigates LED flicker and motion artifacts by implementing a repeated or a random shutter sequence to suppress motion artifacts and decouple exposure from the pulsed light source. - Utilized a 4-tap CIS to extend dynamic range in three discrete steps, maintaining a transition SNR >30 dB across the full range for signal fidelity in automotive applications.	
Ambient-Light-Robust Motion-Resistant Biomedical Imaging System Development	2021 – 2023
- Improved ambient light tolerance by >10x in clinical environments by engineering an 8-tap CIS system that performs quantitative tissue imaging synchronized with pulsed illumination. - Led the LSI design and full tape-out process of a multi-aperture 4-tap CIS, resulting in a successfully fabricated prototype (2024) for non-invasive multi-spectral biomedical imaging.	

PUBLICATIONS

1. Performance Characterization and Tuning of a Charge-Splitting High Dynamic Range 4-Tap CMOS Image Sensor

2025.11

Yu Feng, et al., *Sensors*, DOI: 10.3390/s25226953

2. Programmable Dynamic Range HDR Imaging with LED-Flicker and Motion Artifact Mitigation Using a Four-Tap CMOS Image Sensor

2025.03

Yu Feng, et al., *IEEE Sensors Journal*, DOI: 10.1109/JSEN.2025.3557801

3. Motion-Resistant Three-Wavelength Spatial Frequency Domain Imaging System with Ambient Light Suppression Using an 8-Tap CMOS Image Sensor

2024.01

Yu Feng, et al., *Journal of Biomedical Optics*, DOI: 10.1117/1.JBO.29.1.016006

4. Spatial Frequency Domain Imaging System Using a Scanning Micro-Mirror

2025.03

Kenta Nakazawa, Yu Feng, et al., *Sensors and Actuators A: Physical*, DOI: 10.1016/j.sna.2025.116421

5. Resolving Multi-Path Interference in Compressive Time-of-Flight Depth Imaging with a Multi-Tap Macro-Pixel Computational CMOS Image Sensor

2023.01

Horio Masaya, Yu Feng, et al., *Sensors*, DOI: 10.3390/s22072442

RECENT INTERNATIONAL CONFERENCE PRESENTATIONS

1. Room-Light Operation of a Three-Wavelength Spatial Frequency Domain Imaging System Using Pulsed Illumination and an 8-Tap CMOS Image Sensor

2025.06

Yu Feng, et al., *European Conference on Biomedical Optics 2025*, Munich, Germany.

2. Programmable Dynamic Range Extension up to 110 dB Based on Charge-Splitting Method with 4-Tap CMOS Image Sensor

2025.06

Yu Feng, et al., *International Image Sensor Workshop 2025*, Hyogo, Japan.

3. Multi-Tap CMOS Image Sensor with Programmable Functional Exposure: Application to Structured Light Based Quantitative Tissue Imaging

2024.07

Yu Feng, et al., *Optica Imaging Congress 2024*, Toulouse, France.

INDUSTRY EXPERIENCE

QA Engineer, Meidensha | Nagoya, Japan

2020.04 – 2021.09

- Ensured reliability of EV motor production lines by implementing rigorous quality assurance protocols and defect analysis, reducing line stoppages.

AWARDS & HONORS

Graduate School Scholarship, Amano Foundation

2023 – 2026

Outstanding Academic Record, Shizuoka University

2024 – 2026